



Tohura Shehreen

Contact



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<https://github.com/tohura-shehreen>



About Me

Computer Science graduate with hands-on experience in AI/ML research, full-stack web development, and data-driven problem solving. Built NLP systems for low-resource languages, ML pipelines for bias correction, 3D game engines, and collaborative web platforms. Passionate about applying AI to real-world and societal challenges.

Skills

Technical Skills:

- **Programming:** Python, Java, C, JavaScript
- **ML / AI:** PyTorch, scikit-learn, XGBoost, GNNs, Transformers
- **Web Development:** Flask, HTML, CSS, Bootstrap, JavaScript, MySQL
- **Mobile Development:** Flutter, Firebase
- **Data Handling:** Preprocessing, feature engineering, visualization, analysis
- **Tools / Platforms:** Git, Google Colab, VS Code

Soft Skills:

- Strong communication and presentation skills
- Team collaboration and multidisciplinary project experience
- Participation in community initiatives (Red Crescent, Peace Café Club)

Languages:

- **English:** C1 (IELTS 7.5)
- **Bangla:** Native Speaker
- **French:** Basic (learning)
- **German:** Basic (learning)



Education



B.Sc. in Computer Science

June 2026

Brac University

- **CGPA:** 3.35 / 4.00
- **Key Courses:** AI, ML, NLP, Pattern Recognition, Bioinformatics, Database Systems, System Analysis & Design
- **Thesis:** Neural Machine Translation for Chakma-Bangla: An Exploration of Low-Resource Techniques for Model Optimization



High School Certificate (Science)

2021

Ideal School and College, Motijheel

- **CGPA:** 5.00 / 5.00



Secondary School Certificate (Science)

2019

Willes Little Flower School and College

- **CGPA:** 5.00 / 5.00



Projects



StudyCat — Full-Stack Web App

Ongoing

Solo Project

- Solo-built a full-stack productivity web app: pixel art, Figma UI/UX, React + Tailwind frontend
- Shipped focus timer, session stats, and animated pixel-pet system with outfit/room customization
- Owned entire pipeline — concept, art, code, deployment (Vercel)



Neural Machine Translation for Chakma-Bangla

Group Thesis

- Fine-tuned BanglaT5 via Iterative Back-Translation for extreme low-resource NMT
- BLEU 19.00 (+30.4% over baseline) — matched GPT-4.1 ICL, beat all prior fine-tuned systems
- Python · PyTorch · HuggingFace · BanglaT5 · sacrebleu
- GitHub: github.com/tohura-shehreen/Neural-Machine-Translation-for-Chakma-to-Bangla



StudyHive - a Learning Platform

Group Project

- Led frontend: dashboard design + Flask/MySQL backend integration
- Built YouTube-style resource-sharing hub for students
- GitHub: github.com/tohura-shehreen/studyhive-a-learning-platform-main



3D Maze Escape Game

Group Project

- Built real-time 3D maze game from scratch (PyOpenGL, GLUT, GLU)
- First/third-person cam, enemy AI, collision detection, timed gameplay
- GitHub: github.com/tohura-shehreen/3D-Maze-Escape-Game



eBird Bias Correction (GNN + XGBoost)

Solo Project

- Built ML pipeline correcting bias in citizen-collected bird data (GNN + XGBoost)
- Owned preprocessing, modeling, evaluation, and documentation end-to-end
- GitHub: <https://github.com/tohurashehreen/ebird-bias-correction-using-gnn-and-xgboost>



Experience



Small Business Owner — Online Clothing Shop

Aug 2025 - Present

- Independently managed end-to-end operations including social media marketing, paid ad campaigns, order fulfillment, and customer communication
- Grew social media following through organic content strategy and targeted advertising



Tutor — School Students

2023-2025

- Tutored primary and middle school students, breaking down complex concepts and adapting teaching methods to individual learning styles